

Part I: Write “True” if the statement is correct and “False” if the statement is incorrect (1pt each = 5pts)

1. Direct array-to-array assignment are allowed in C++, making it easy to copy the contents of one array into another.
2. In for loops, the body of the loop is executed before the condition is tested to determine whether it is true or false.
3. Struct a (struct int a;} is a proper definition of structure.
4. Reference operator is used to get the value stored in a computer memory from a given address.
5. Functions, just like they can accept arrays as parameters, they can also return arrays.

Part II: Choose the best answer among the given alternatives (1pt each = 15pts)

1. Which control statement is suitable for situations where a block of code needs to be executed at least once?
A. do-while loop C. While loop
B. For loop D. Switch statement
2. Which of the following string declaration is incorrect?
A. char str[10]; C. string str;
B. char str[] = "hello"; D. [uncertain — partially obscured in source]
3. Which of the following statements is the correct way of using a pointer?
int x=5; int *p;
A. p=x; C. *p=x;
B. p=&x; D. *p=&x;
4. Which for-loop statement will iterate through all elements of an array where N is the size of the array?
A. for(int i=0; i<N; i++)
B. for(int i=array[0]; i< array[N]; i++)
C. for(int i = 0; i < array[N]; i += 1)
D. for(int i = 0; i < N; i +=1)
5. What is the output of the following code?

```
char str1[50]="Jimma is a big town";  
char str2[]="Moon revolves around the earth";  
strncat(str1, str2, 13);  
str1[24]='\0';  
str2[13]='\0';  
cout<<str1<<" "<<str2;
```


A. Jimma is a big town Moon. Moon revolves
B. Jimma is a big town Moon. Moon revolves around the earth
C. Jimma is a big town Moon revolves. Moon revolves around the earth
D. Moon revolves around the earth. Moon revolves
6. [Question 6 — partially obscured in source image]
7. [Question 7 — partially obscured in source image]

8. What is the output of the following code?

```
int i, j=0;
for(i = 33; i >= 3; i -= 7)
{
    if ( (i % 2) || (i / 5 > 2) )
        j += i;
    cout << j;
}
```

A. 67 B. 40 C. 83 D. 92

9. [Question 9 — partially obscured in source image]

10. How many times does the following loop execute?

```
for(int i = 20; i >= 0; i -= 3)
    cout<<i;
```

A. 20 B. 17 C. 0 D. 7 E. 15

11. [Question 11 — partially obscured in source image]

12. Which of the following statements is true regarding the switch statement compared to the if..else statement in programming?

- A. The switch statement is more suitable for complex logical conditions.
- B. The if..else statement allows for multiple conditions to be checked in a single statement.
- C. The switch statement can be used to compare floating-point numbers more efficiently.
- D. Comparison operations other than equality can be handled in the switch case.

13. What will be the output of the following CPP code fragment?

```
char str [] = "password";
if(strcmp (str, "password"))
{
    Cout<<"success";
}
else
{
    Cout<<"Incorrect";
}
```

A. Run-time Error C. Incorrect
B. Success D. No output

14. The keyword used to transfer control from a function back to the calling function is:

A. goto C. Return
B. Switch D. Exit

15. Which one of the following is false?

- A. When arrays are passed as parameters, they are always passed by reference.
- B. A function can return only a single value.
- C. In pass by reference, changes made inside a function have no effect on the argument.
- D. Pass by value copies the values stored in arguments in to the parameters.
- E. void function is a function that does not accept any argument.

Faculty of Electrical and Computer Engineering
Computer Programming Final Exam for Pre-Engineering Students

Course Code: FECEg4202

Date: January 12, 2024

Time Allowed: 2 hours

Max Pts.: 40%

Academic Year: 2016 E.C

Instruction:

- Switch off your mobile
 - Check the question paper has 5 parts on four pages
 - Answer on the provided space only
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PART 1: CHOICE (1.5 point each)

Choose the correct answer among the given alternatives

1. Choose the type of loop which is guaranteed to execute at least once?
A. for loop C. while
B. do-while D. none
2. Selective switch is equivalent to:
A. nested if-else C. do-while loop
B. for loop D. if statement
3. The function that calls itself is known as:
A. Prototype C. Recursion
B. Inline D. A and B
4. Which of the following correctly declares an array?
A. int array1 [10]; C. array1;
B. int array1; D. array array [10];
5. Where does the execution of the program start?
A. user-defined function C. main function
B. void function D. none of the mentioned
6. What is the correct definition of an array?
A. An array is a series of elements of the same type in contiguous memory locations
B. An array is a series of element D. An array is an element of the different type
C. An array is a series of elements of the same type placed in non-contiguous memory locations
7. Choose the right option. string *x, y;
A. x is a pointer to a string; y is a string. C. both x and y are pointers to string types
B. y is a pointer to a string, x is a string D. y is a pointer to a string
8. Why is a pointer to a pointer needed?
A. To hold address of variable C. To access the value of pointer
B. To hold address of pointer D. All
9. Decision Control statements in C++ can be implemented using:
A. If C. Conditional Operator
B. if-else D. All of the above

10. If you have to make a decision based on multiple choices, which of the following is best suited?
- A. If C. switch
B. if-else D. All of the above

PART 2: TRUE OR FALSE (2 point each)

Write T if the statement is correct or F if not correct statement working

1. Assume `char a[]="Hello"; char b[]="hello"`. The output of `strcmp(a, b)=0`.
2. Total sum of elements in integer array can be calculated by: `int sum=0; for(int i=0; i<10; i++) cout<<sum+=x[i];`
3. Given the declaration: `int data [10]; int *p;` then the statement: `p = data;` is invalid in C++.
4. A function can be called by any other functions.

PART 3: MATCH (1.5 point each)

Section 1: Match the statement under Column A with corresponding output under Column B.

Suppose: `Char *n; char a[] = {'A','B','C','D'}; n = &a[2];` statements are in main function.

Assume the program is executed sequentially (1, 2, 3, 4, and 5).

Column A

1. `cout<<n;`
2. `cout<<*n;`
3. `cout<<--n;`
4. `cout<<*(++n);`
5. `cout<<++n;`

Column B

- A. gives address of 'B'
- B. gives 'C'
- C. gives 'D'
- D. gives address of 'C'
- E. gives address of 'D'

PART 4: Fill Blank Space (2pts each)

1. Because cin don't support tabular, blank and space we use _____ instead of.
2. _____ loop is not recommended to implement it in a place where it is most needed, which can lead to the loss of human life by one mistake.

PART 5: Work Out (6pts)

1. Write a C++ program that displays the days of a month from the 1st day up to 30th day sequentially. Below the loop body make it display your full name. Write the program as well as the output of the program clearly. (4pts)
 - a. At the 15th, and 17th day, the program calls the continue, and break respectively.
 - b. At the 15th, and 20th day, the program calls the continue, and exit() function respectively.
7. Write a program that selects either +, -, /, * and unknown if different from those based on the user's input with switch. (4pts)

Name: _____ Id: _____

Instructions:

- Any form of cheating results in an immediate disposal of the candidate
 - Answer each question only the space provided
 - Tidy and neat works earn you extra marks
 - Make sure the exam has 2 pages and 7 questions
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1. To call a function it must have been declared previously. This is called _____ (2pts)
2. _____ use the same function name for more than one function. (2pts)
3. With cstdlib (stdlib.h) library **atoi** is converting from _____ to _____, **atol** converting _____ to _____, and **atof** _____ to _____. (2pts)
4. In Function for string manipulation, what is? (4pts)
 - A. strcat: _____
 - B. strcmp: _____
 - C. strcpy: _____
 - D. strlen: _____